Abstract

Computational Fluid Dynamics (CFD) is often taught with pre-built solvers that rely on complex, already implemented, numerical methods and data optimization schemes. Pre-built solvers can be expensive to run and purchase, limiting the opportunities students have to experiment with their own simulations. This hinders the ability for students to understand and experiment with simple accessible solvers to gain a more dynamic understanding of CFD. With the power that CFD holds for the future of engineering it is crucial to provide as many open resources for future engineers. In an effort to simplify the basic logic and schemes intended for use in CFD this paper aims to provide an insight into the Finite Volume Method (FVM) and why it is the preferred method for CFD solvers. Our goal is to understand, set up, and analyze the potential flow over airfoils in 3 core steps; building the governing equations for the system, discretizing the geometry of the volume and the equations for simulation set up, converging the data to a meaningful result then providing plots for students to visualize these concepts and the potential flow over certain airfoils. The explanation of a topic will be broken into 2 pieces, an initial informal approach to guide basic sense of the topic and a formal approach to provide the clear construct into the mathematics behind simulation to guide students from intuition to implementation. The key final fundamental learning is to develop a clear understanding of how, why, and when to use geometric discretization to set up a simulation, and how to interpret and utilize the visual and numerical data provided by the simulation to further an understanding of airfoil theory.

Derivation of the Governing Equations

The basic building blocks behind any physics analysis are conservation laws, properties of matter cannot be created or destroyed. Rather than beginning with trying to understand the conservation of millions of particles in a fluid moving through some volume we can begin intuitively with just one little particle moving around. Our particle is the only thing to exist in its universe, so the entire space in which this particle can move through only contains the particle itself. It's a simple statement but a powerful one, if we say this universe extends out to infinity, this becomes our unbounded system that our particle can't leave, then the total amount of mass, energy, anything within this system is based on the properties of this lone particle. The particle is also moving around so it does have some velocity, and its position isn't the same each time you look at it but it still remains in our universe. It isn't far from reason to say that this particle isn't going to pop away from existence or that another particle will spawn into this universe somewhere. So no matter what this universe will always contain the same properties that of our particle for ever in time no matter where the particle is. That's basically our conservation laws, the mass of the particle is the mass of the system and no matter what that is always going to be the mass, same goes for energy and any other properties that go with that particle even if the particle is moving around.

Lets define our universe an infinite empty volume which extends in every direction, thus the volume can be defined spatially as $V = xyz$. Within this volume there exists our single particle χ which has some mass m and has an incredibly small finite volume. It can be stated that χ contains scalar properties of matter which we will call ξ. If χ is the only known matter to exist in this volume, logically the total amount of χ within V is $m\xi$. Imagine if χ was moving at some velocity u in the x-direction, regardless of the velocity of χ we understand that $m\xi$ will remain consistent in V. Since V stretches out infinitely, analyzing a small control space, CV, which intersects the path of a moving particle can allow for further understanding about how moving matter must be conserved within our system. Let us say χ enters CV at some point $\alpha(0, y, z)$ at $t = 0$ and exits at point $\beta(x, y, z)$ at some time $t > 0$. We want to see the total amount of $m\xi$ moving through CV. To visualize the context of this situation mathematically we can integrate $m\xi$ along its path from α to β,

$$\int_{\alpha}^{\beta} m\xi dx.$$

χ must have some velocity, otherwise χ would not be able to transport itself from α to β, hence we can transform our integral from one dependent of space to time using the relationship $dx = udt$ where $\alpha \rightarrow t = 0$, and $\beta \rightarrow t > 0$,

$$\int_{0}^{t} mu\xi dt,$$

thus,

$$mu\xi_{t} - mu\xi_{0}.$$

Since χ's position is some function of time $\chi(t)$ and $\chi(t) = \chi(t0)$ at all times, than $mu\xi(t2) - mu\xi(t1) = 0$. Hence we can say that any changes or fluctuations of $\chi(t)$ within some control space dV must be 0,

$$\frac{\Delta(mu\xi)}{\Delta t} = 0.$$

Taking the limit as $\Delta t \rightarrow 0$ our equation transforms into a derivative,

$$\frac{d(mu\xi)}{dt} = 0$$

Any scalar properties of χ will be conserved even if m, ξ, and u fluctuate, the total change of these quantities is dependent on $\dot{m}$. Thus the equation below fully satisfies the condition the matter and properties of it are conserved for all times t, within our system CV thus satisfying the case for our entire volume,

Above we have denoted the conservation of χ in our entire volume, saying that as our particle's properties as it moves in a straight path with some velocity will always be conserved inside our universe at any given time, so that value never changes ever in the universe. An

important issue arise though, we have only described the particle moving linearly in the x-direction. We want a way to describe the particle moving in every possible direction since our volume is 3-dimensional. We can argue that we have proved that as long as the particle is moving linearly in any direction properties of the particle have still been conserved, even if we only analyzed the movement in the x-direction we can say regardless of the direction the path χ will still enter the control volume at some point and leave out the other over some period of time, satisfying conservation laws. So if the particle is moving in an exact straight line then we can concretely say we have described its conservation, but our particle can wander its path does not even have to be linear and it can be bounded, the particle could be rotating around some point so its path loops back around to where it started. We need to satisfy that no matter whatever way our particle chooses to move we can still mathematically describe its motion and its conservation within a system.

Considering the case above, imagine ū as some vector whose origin is at the center of χ and models the magnitude and direction of χ's velocity. If χ moves in the x-direction, than $\bar{u} = ui + 0j + 0k$, where u is the magnitude of velocity. If χ can move at any velocity in any direction freely throughout V, than $\bar{u} = ui + vj + wk$, where u, v, w are the velocity magnitudes for their respective directions. Since χ can move freely than its position and velocity can all be considered dependent on time, $u(t)$, $v(t)$, $w(t)$.

$$\bar{u}_{\chi} = u(t)i + v(t)j + w(t)k$$

We had previously derived the conservation equation for χ moving in some linear direction, but the path doesn't need to be exactly described as linear. As long as the path of χ never intersects a position it had been in before then it must always exit a control volume if it enters it. Hence if the path of χ remains in an unbounded than it must leave the control volume after some amount of time has passed if $\bar{u} > 0$. So χ's path of travel does not need to remain perfectly linear as long as the path remains unbounded. The issue now arises about what happens if that path is bounded. Imagine a spherical control volume dS with a radius R and a center at $(0, 0, 0)$, now χ orbits around the center of dS on a closed curve with some velocity $\bar{u} = u(t)i + v(t)j$ at some distance r away from the center where $R > r$. Thus going against the argument as long as $|\bar{u}| > 0$ for χ, as $t \to \infty$ χ must exit dV, but the total amount of $m\xi$ in V still remains $m\xi$ with the case above, χ's path still satisfies the conservation of mass. For each small step in time forward dt, $\chi(t)$ will take some step forward along its path df. The length of a path is $L = \oint ds$, since χ's position is dependent on time we can say,

$$ds = \frac{df}{dt}dt.$$

The integral of the path now becomes $L = \oint \bar{u}dt$ where $df = \bar{u}dt$. If we consider the movement of χ on the path we see the equation become, $\oint \Omega\bar{u}dt$. Since χ can be described as a combination scalar characteristics of mass, the total amount of χ moving along the curve is, $\oiint m\xi\bar{u}dt$ Since $\chi(t2) = \chi(t1)$ at all times, $m\xi(t2) - m\xi(t1) = 0$; thus proving that even if χ moves along some closed contour the total amount of χ in a system is conserved as well as moving in an unbounded direction. Thus the conservation equation can be written as,

$$\frac{d(m\bar{u}\xi)}{dt} = 0$$

Now that we have described how the particle itself moves about in our universe and how it is inherently conserved within that system whether in motion or stationary. The question arises as to what causes the particle to move initially, or what could displace it off its current path? The first Law of Motion dictates any object at rest will remain at rest until some force acts on it, and or any object in motion will remain in motion at a constant speed in a straight path until some force acts upon it. If we reconsider the case of our particle moving linearly in the x-direction and any changes in velocities of our particle are also along the x-direction then we can infer that the direction of the forces required to change the particle's velocity are parallel to the direction it travels. An important variable to consider is the mass of the particle, the heavier the particle the more force we would need to accelerate the particle to a certain rate and the opposite for a lighter particle, or better known as the second Law of Motion.

As χ is in motion, we understand it will remain until some force acts upon it due to the inertia of χ. The net forces acting on χ will be 0 if the velocity of χ is constant. Imagine some constant force F acting on χ, so u must change at some rate proportional to the scalar F. As the force acting on χ increases, the change in velocity or acceleration, a or $\Delta u/\Delta t$, will behave similarly, likewise as the force on χ decreases, $\Delta u/\Delta t$ decreases thus $F = a$. The rate at which an object's velocity changes when the force acting upon it is inversely proportional to the object's mass. If χ's mass increases than the acceleration of χ would decrease, it would require more force to push a more massive object, and less force for an object with less mass. Hence the motion of χ can be quantified by its momentum, $p = mu$, so the forces acting on χ is equivalent to the flux of χ 's momentum,

$$F = \frac{d(mu)}{dt}.$$

We have described the force required to move our particle at different speeds in one dimension. To be noted as this can be applied to the y and z directions it doesn't yet express the cohesive interactions for all three dimensions. To describe motion in 3 dimensions we need to articulate the forces acting on the body of our particle. We can simplify the geometry of the particle to a cube with six faces, two faces pointed in the x-direction (left and right), two faces faces towards the y-direction (up and down), and two faces to the z-direction (forward and rear). We can look solely at understanding the motion along the one direction, let's consider are particle accelerates in the x-direction so any forces acting on the particle will be on one of its two x faces. One of the x faces will look towards the negative x-direction (leftward) and the other in the positive x -direction (rightward). If the particle accelerates along in the positive x-direction, or towards the right, we expect some force to push on the left-hand face of the particle in that direction. Now let's say an equal force pushes the right-hand face of the particle, though there are no changes in the velocity, our particle still experiences a constant inward force acting on it. This inward force or pressure can be said to be experienced on every face of our particle, we can simplify and say that when our particle experiences no changes in velocity this inwards pressure is zero for all the collective faces. Now if the particle experiences acceleration we can say one of the faces experiences a greater force than the force being applied to the face opposite of it.

The amount of work done on Ω is proportional to the external forces on Ω over some distance. Imagine a one dimensional case where the total work done on a system is, $W = \int F dx$. We can substitute for F using the momentum equation where, $F = ma$, thus $W = \int ma dx$. We understand that $a = du/dt$, and $dx = udt$ due to $u = dx/dt$, hence,

$$W = \int m \frac{du}{dt} u dt.$$

The dt will cancel out leaving us with $W = \int mudu$; solving the integral,

$$W = \frac{1}{2} m u^2.$$

We see the work done on Ω is related to the change kinetic energy, $\int mudu \rightarrow KE2 - KE1 = W = \Delta KE$. An important relationship when considering the energy of systems made up of many particles. We see that this is from some initial position where Ω is at rest,

$$W = \int_0^u mudu.$$

In the case of a closed system, where $W = \Delta KE$ and $\Delta IE = Q - W$, IE is the internal energy of an object, we see $Q = \Delta KE + \Delta IE$ or $Q = dKE + dIE$. The total energy TE of a system is the sum of the Kinetic, Internal, and Potential energies of an object, $TE = KE + IE + PE$. If Ω is the only matter in V than it makes sense to assume no potential field, $PE = 0$, such now $TE = KE + IE$. The total change in energy of a system should remain 0, $dTE = 0$. If $Q = \Delta KE + \Delta IE$ than $TE = Q$. So,

$$dQ = 0$$

So far we have considered the motion of Ω in an environment where it is the only matter in the

A fluid flow is the motion of particles through a fixed control volume we choose to study. These particles are assumed to be infinitesimally small but contain properties such as mass, position, volume, temperature, etc. Imagine an infinite empty volume in which exists a single particle Θ with some mass m. If Θ is the only known matter to exist in this volume, logically the total mass within the volume at $t = 0$ is equivalent to mass at anytime when $t > 0$,

$$m_\Theta(t_0) = m_\Theta(t_n).$$

Inherently we know that mass must be conserved at all times, to satisfy this argument we can rearrange our equation where all terms of mass are on one side,

$$m_\Theta(t_n - t_0) = 0.$$

We see there exists no fluctuations of the total mass within this volume. Considering that this exists for single particles and conservation laws state that mass cannot be created or destroyed in a system, or mass must be conserved in a closed system, a system made of many many particles must also conserve the total mass. If each particle contains some mass, then the total amount of particles in a specific volume is the density of that fluid. Therefore when considering a fluid, the total mass of the fluid can thus be found by taking the volume integral of the fluid density, $m = \iiint \rho dV$. We can relate this by saying the rate at which mass enters a volume is proportional to the mass flow rate, or the mass flux entering a volume, where ū is the velocity of the fluid flow,

$$\frac{\partial}{\partial t} \iiint \rho dV = - \oiint (\rho \bar{u} \cdot n) dA.$$

We can rearrange the equation to state that the rate at which we have mass entering and the flux leaving a system's sum is 0. Using Gauss's Theorem to transform our surface integral to a volume integral, $- \oiint (\rho \bar{u} \cdot n) dA = - \iiint (\nabla \cdot \rho \bar{u}) dV$, transforming the relationship to,

$$\iiint [\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \bar{u})] dV = 0.$$

If $\int f(x) dx = 0$ than $f(x) = 0$, it can naturally be assumed the conservation equation can be simplified to,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \bar{u}) = 0.$$

Expand $\nabla \cdot (\rho \bar{u})$, obtaining $\nabla \cdot (\rho \bar{u}) = \rho(\nabla \cdot \bar{u}) + \bar{u} \cdot (\nabla \rho)$. Where the left hand term, $\rho(\nabla \cdot \bar{u})$, is the expansion or compression term. The density of the fluid at some point is relative to how much fluid flow diverges or converges at that point. As the divergence is greater than 0, we expect density to get smaller, similarly as divergence is less than 0, we expect density to increase. The right hand term, $\bar{u} \cdot (\nabla \rho)$, is the convective transport of density. Where $\nabla \rho$ are the variations in density across the fluid. The convective transport of these variations is simply how fluid of different density moves around the system. When the convective term is either greater or less than 0 the density of the fluid moving into or out of the region is either denser or not, ie. increasing or decreasing the local density. Thus the equation of continuity finally becomes,

$$\frac{\partial \rho}{\partial t} + \rho(\nabla \cdot \bar{u}) + \bar{u} \cdot (\nabla \rho) = 0.$$

As this fluid is in motion, we understand it will remain until some force acts upon it due to the inertia of the fluid. Θ will remain at rest or in motion until some force acts on it. Hence the net forces acting on Θ will be 0 if the velocity of Θ is constant. It is understood the acceleration of an object is directly proportional to the forces being applied and inversely proportional to the object's mass, otherwise stated as the sum of forces acting on an object is equivalent to the flux of momentum,

$$F = \frac{d}{dt}(mu).$$

When considering Θ motion, the momentum equation is sufficient to describe the relationship between the momentum flux and the forces acting on the particle. Since the mass of the particle remains constant, for this reason any change in mass throughout time must be 0, $\Delta m = 0$,

$$F = m_{\Theta}\frac{du}{dt}$$

A fluid volume is a continuous system where characteristics of the system, such as momentum, can be carried through the flow. As we analyze this flow moving through some control volume we must account for variations over time of these characteristics at fixed points as well as accounting for how these characteristics are carried through the flow, convective transport. Suppose some scalar quantity ξ that is dependent with space and time, $\xi(x, y, z, t)$. The total change in ξ over some interval in time Δt is, $\Delta\xi = \xi(x + \Delta x, y + \Delta y, z + \Delta z, t + \Delta t) - \xi(x, y, z, t)$. Using a Taylor Series Expansion and neglecting higher order terms,

$$\Delta\xi \approx \frac{\partial\xi}{\partial x}\Delta x + \frac{\partial\xi}{\partial y}\Delta y + \frac{\partial\xi}{\partial z}\Delta z + \frac{\partial\xi}{\partial t}\Delta t.$$

Dividing both sides by Δt,

$$\frac{\Delta\xi}{\Delta t} \approx \frac{\partial\xi}{\partial t} + \frac{\partial\xi}{\partial x}\frac{\Delta x}{\Delta t} + \frac{\partial\xi}{\partial y}\frac{\Delta y}{\Delta t} + \frac{\partial\xi}{\partial z}\frac{\Delta z}{\Delta t}.$$

Finally taking the limit as Δt approaches 0 we see the nature of this derivative,

$$\frac{D\xi}{Dt} = \lim_{\Delta t \to 0}\frac{\Delta\xi}{\Delta t} = \frac{\partial\xi}{\partial t} + \frac{\partial\xi}{\partial x}\frac{\Delta x}{\Delta t} + \frac{\partial\xi}{\partial y}\frac{\Delta y}{\Delta t} + \frac{\partial\xi}{\partial z}\frac{\Delta z}{\Delta t}.$$

We see,

$$\frac{D\xi}{Dt} = \frac{\partial\xi}{\partial t} + \bar{u} \cdot (\nabla\xi).$$

The derivative obtained is the material derivative. It satisfies the intended demand that governing equations must contain both local and convective transformations. If ξ is some property of a fluid it also must be of some unit mass, therefore we see the total amount of ξ in some specific volume of fluid is $\iiint \rho\xi dV$. The local fluctuations of ξ are,

$$\frac{\partial}{\partial t}\iiint \rho\xi dV.$$

We can consider the convective term above the flux of $\xi\bar{u}$ through some surface, thus the flux through some surface be written as $\oiint(\rho\xi\bar{u} \cdot n)dA$. Hence,

$$\frac{D}{Dt}\iiint_{System}\rho\xi dV = \frac{\partial}{\partial t}\iiint_{CV}\rho\xi dV + \oiint_{CS}\rho\xi(\bar{u} \cdot n)dA,$$

better known as the Reynolds Transport Theorem. Now the transport of momentum becomes, where $\xi = \bar{u}$, we can rewrite $\Sigma F = ma$ using this relationship to derive an equation of motion,

$$\Sigma F = \frac{D}{Dt}\iiint \rho\bar{u}dV$$

Where $\frac{D}{Dt}\iiint \rho\bar{u}dV$ can be solved using Reynolds Transport Theorem where,

$$\frac{D}{Dt}\iiint_{System}\rho\bar{u}dV = \frac{\partial}{\partial t}\iiint_{CV}\rho\bar{u}dV + \oiint_{CS}\rho\bar{u}(\bar{u} \cdot n)dA.$$

Thus the final equation is,

$$\Sigma F = \iiint[\rho\frac{\partial\bar{u}}{\partial t} + \bar{u}\frac{\partial\rho}{\partial t} + \rho(\bar{u} \cdot \nabla)\bar{u} + \bar{u}(\bar{u} \cdot \nabla\rho) + (\bar{u} \cdot \nabla)\rho\bar{u}]dV$$

Finite Volume Method

Take some closed solid body model with some surface, $S \subset \mathbb{R}^3$. If ζ is some larger boundary that encases the model, let $\partial\zeta \subset \mathbb{R}^3$ be the boundary of the outer domain. The volume V, inbetween S and $\partial\zeta$ will be the control volume where flow analysis will occur. This set control volume gives a manageable space for discretization; say V is a composition of tetrahedral cell ς containing some little volume dV where $\varsigma_i \cap \varsigma_j = \emptyset$ for $i \neq j$,

$$V = \bigcup_{G\in} \varsigma_k.$$

The discretization of the volume has separated it into a union of non overlapping cells. Begin with giving further definition to ς. Each cell is comprised of a set of four vertices, which we will list $a(x, y, z)$, $b(x, y, z)$, $c(x, y, z)$, and $d(x, y, z)$. If each cell structure remains unchanged for the duration of the simulation then certain corresponding characteristics/properties will remain unchanged. At the center of each cell we will define a point referred to as a centroid, G. The centroids positions can be found using its relation to its unique 4 vertices, $G(x, y, z)$ will be defined as the center of ς. Every ς in V must have a G, hence we can define a ς by its G,

$$V = \bigcup_{k=1}^{K} G_k.$$

Each cell is also made up of a set of faces, f, in the particular case we are studying currently where cells are geometrically clarified as a tetrahedra; it is understood that G is composed of 4 faces with each its unique area based on its vertices, ie. $f1 = \triangle abc$, $f2 = \triangle abd$, $f3 = \triangle acd$, and $f4 = \triangle bcd$. Each face has some area A, and the sum of A across the total number of faces will be the surface area of G. Using the Cross Product Method we can calculate the faces of each cell. For example $f = \triangle abc$, than $A = 1/2||AB \times AC||$, where vector $AB = b - a$ and vector $AC = c - a$. Each face has a midpoint, which can simply be calculated using the vertices of the given face, and some outward facing normal vector n protruding from that midpoint. To compute the normal vector of the face consider once again, $f = \triangle abc$, $AB = b - a$, $AC = c - a$, and the vertice opposite the face, in this case d. Say $N = AB \times AC$, than:

$$n = \frac{N}{||N||}.$$

The totality of the control volume V will be made by a set number of centroids G. Each G will have four faces, four outward facing normals, and occupies some volume dV. Consider G is possible of holding values of data such as ρ, ū, ξ, etc; any flow through V must encounter G, thus local properties and data of the flow can be stored in G.

Uniform Potential Flow with FVM

V contains some model that occupies its spaces, for now see that V is free of such a model. Let us observe some fluid flow within this control volume, imagine a steady uniform flow of an incompressible, irrotational, inviscid fluid flowing within V moving in the x-direction; the magnitude or direction in x is yet to be determined. In an empty domain, free of any designs, discretized into small identical cells, each cell comprised of some centroid G, should hold the same value for ū considering the flow is uniform. We have defined the basic characteristics of a flow as its mass, velocity, and some scalar property inherent to the flow. Across the boundary domain the total fluctuations of these characteristics should remain 0. Key points of continuity so far established are mass, momentum, and energy; evaluating the total mass flux of a fluid,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \bar{u}) = 0.$$

Since $\nabla \rho = 0$, it can be assumed that total density of the fluid will remain constant throughout V, hence why every G in V will all contain identical values of ρ, and simplifying the continuity equation to, $\nabla \cdot \bar{u} = 0$. The total divergence, outward flux, of ū through V can be thought of intuitively as the summing up of local divergences of ū at points within V. Instead of an infinite set of points in V, rather examine and sum up the local divergence of ū at every $G(x, y, z)$, where the divergence within the cell volume must still abide by conservation laws, $\iiint (\nabla \cdot \bar{u}) dV = 0$. Since $\oiint (\bar{u} \cdot n) dA = \iiint (\nabla \cdot \bar{u}) dV$, by Gauss's Theorem, the total sum of fluxes through G within V must also remain 0,

$$\sum_{k=1}^{K} \oiint_{\partial G_k} (\bar{u} \cdot n) dA = 0.$$

Hence fluid flowing through V can be computed by analyzing local flux's at all G, since the sum of the internal fluxes collapses, we are left with $\oiint_{\partial \varsigma} (\bar{u} \cdot n) dA = 0$, satisfying continuity conditions for the boundary domain.

If $\oiint (\bar{u} \cdot n) dA = 0$ must hold true at every point G, and since G is composed of four faces with unique areas and four outward facing normals. We can approximate the flux through the faces as ū dotted with the normal vector unique to that face multiplied by the faces area, $\Phi \approx \bar{u} \cdot nA$. For $\oiint (\bar{u} \cdot n) dA = 0$ to hold true, the total sum of fluxes through the faces that make up G must remain 0, $\sum \Phi_f = 0$, or,

$$\sum_{f\in\partial G} \bar{u} \cdot n_f A_f = 0.$$

We see that $\oiint_{\partial G} (\bar{u} \cdot n) dA \approx \sum_{f\in\partial G} \bar{u} \cdot n_f A_f$ or $\sum_{f\in\partial G} \Phi_f$, and for $\oiint_{\partial \varsigma} (\bar{u} \cdot n) dA = 0$ to remain satisfied we expect the discretized equation to be,

$$\sum_{k=1}^{K} \sum_{f\in\partial G_k} \Phi_f = 0.$$

Imagine some centroid G within V, since V can be thought of as a collection of cells we can assume G will have some neighbors we will call N. The cell is a tetrahedral, we can assume G will have a neighboring centroid that shares on of its faces, thus G will have four neighboring centroids, $N1$, $N2$, $N3$, and $N4$. We can consider this the local neighborhood of G, thus any flux entering G are directly resulted from flow leaving G's neighbors, and vice versa any flow leaving G must enter its local neighborhood, so flux is directly shared between a centroid and its neighbors. A key exception has to be brought to attention, there exist cells that have a face which lies on the outer domain surface. Centroids of these cells do not have neighbors across from this face on the pretence that our analysis is finite. We can imagine than a ghost neighbor to hold its place, an imaginary centroid outside of $\partial\zeta$ mirroring the centroid across the face that they hold common. Fluxes now can be calculated between all the faces of G on $\partial\zeta$ with the help of these ghost neighbors.

It is known that G holds a value of scalar and vector quantities such as $\bar{u}$. But an important distinction arises, if flow is non-viscous, $\nabla \cdot \bar{u} = 0$, and $\nabla \times \bar{u} = 0$ is true than $\bar{u}$ can be the gradient of some scalar field, $\bar{u} = \nabla\varphi$. The gradient of φ at G can be approximated using the FVM scheme and the respective neighborhood of G. The flux of a vector field, f, through some closed surface is, $\iiint(\nabla \cdot f)dV = \oiint(f \cdot n)dA$. Hence the gradient of a scalar, φ, through a closed surface would be, $\iiint(\nabla\varphi)dV = \oiint(\varphi n)dA$, or discretized, $\nabla\varphi V = \sum\varphi n A$. So,

$$\nabla\varphi_G \approx \frac{1}{dV}\sum_{f\in\partial G}\varphi_f n_f A_f$$

If V is some three dimensional volume that has some spacial centerpoint positioned at the origin, than any centroid can be given some positional vector, $\dot{G}$, along with its corresponding neighbors, $\dot{N}$. The straight-line distance, d, separating G and N can be found practically by, $||\dot{N} - \dot{G}||$, giving analytical data for d. The total sum of changes in some scalar by some displacement multiplied by the corresponding directional derivative returns the gradient of that scalar at that point; the directional derivative, $\dot{d}$, for this case can be found with $\dot{N}$ and $\dot{G}$. Since $\bar{u} = \nabla\varphi$, we can fully approximate $\bar{u}$,

$$\bar{u}_G \approx \frac{1}{dV}\sum_{f\in\partial G} A_f \frac{\varphi_{N_f} - \varphi_G}{||\dot{N}_f - \dot{G}||}\dot{d}$$

The total flux of $\bar{u}$ through G, $\Phi \approx \bar{u} \cdot nA$, can be rewritten now that we have a close estimation of $\bar{u}$ within the system of G and its neighbors. If some fluid is flowing through every centroid G within V must have some fluctuation of $\bar{u}$ through its faces, thus conservation laws must hold true for$V$ as a system, $\oiint_{\partial\zeta}(\bar{u} \cdot n)dA = 0$. Than,

$$\sum_{k=1}^{K}\sum_{f\in\partial G_k}\left(\frac{1}{dV}\sum_{f\in\partial G} A_f \frac{\varphi_{N_f} - \varphi_G}{||\dot{N}_f - \dot{G}||}\right)(\dot{d} \cdot n_f)A_f = 0.$$

Considering the fluid progressing in any direction through V, we see that if some uniform flow, ie. the flow's velocity will remain constant throughout the body of fluid; moves through our discretized volume than it can be stated that the local velocity at points G in V will remain constant and identical which we can refer to as, U as some constant velocity of the flow, therefore $\nabla\varphi = U$. Since V is the collection of evenly and identically discretized cells, than the total transfer through V is conserved because $\bar{u} = U$ at all G. $\oiint(U \cdot n)dA = \iiint(\nabla U)dV = 0$, meaning our discretized equation is a valid method for analyzing the uniform flow in any direction through our volume.

Potential Flow Past Bodies with FVM

Say now that V is no longer free of the model it once contained, than once again V becomes the discretized space in between S and $\partial\zeta$; but still some uniform flowing fluid through the system. No fluid can penetrate through S, so $\bar{u} \cdot n = 0$ at the surface of the model. There exist $G\ in\ V$ that share some face that is on S.

Potential Flow

There exists certain vector fields f that are the gradient of some scalar function F; $f = \nabla F$. For $f = \nabla F$ to be true, than $\nabla \times f = 0$. A well known vector calculus identity states the curl of any gradient must equal 0, $\nabla \times \nabla F = 0$. These vector fields are considered gradient flows or irrotational flows. For a potential flow the vector field must additionally be incompressible, $\nabla \cdot f = 0 \rightarrow \nabla \cdot \nabla F = 0$. $\nabla \cdot \nabla F = 0$ is the well known Laplace's Equation :

$$\nabla^2 F = 0$$

Consider a vector field as the velocity of some fluid flow, $\bar{u} = u_i + v_j$. If the fluid flow is incompressible, irrotational, and steady, ie: $\nabla \cdot \bar{u} = 0$, $\nabla \times \bar{u} = 0$, $\frac{\partial\bar{u}}{\partial t} = 0$, then it is a potential flow. Written as $f = \nabla F$, the fluid field is equal to the gradient of some potential function φ:

$$\bar{u} = \nabla\varphi$$

$$\bar{u} = \nabla\varphi = \frac{\partial\varphi}{\partial x}, \frac{\partial\varphi}{\partial y} \quad \text{Cartesian}$$

$$\bar{u} = \nabla\varphi = \frac{\partial\varphi}{\partial r}, \frac{1}{r}\frac{\partial\varphi}{\partial\theta} \quad \text{Polar}$$

Taking the divergence of the potential; $\nabla \cdot \bar{u} \rightarrow \nabla \cdot \nabla\varphi$, the function must satisfy these conditions to be considered potential:

$$\nabla \cdot \nabla\varphi = \frac{\partial^2\varphi}{\partial x^2} + \frac{\partial^2\varphi}{\partial y^2} = 0 \qquad \text{Cartesian}$$
$$\nabla \cdot \nabla\varphi = \frac{\partial^2\varphi}{\partial r^2} + \frac{1}{r}\frac{\partial\varphi}{\partial r} + \frac{1}{r^2}\frac{\partial^2\varphi}{\partial\theta^2} = 0 \qquad \text{Polar}$$
$$\nabla^2\varphi = 0$$

A method to visualize these flow fields is streamline function notated by ψ. These curves are always tangential to our vector field providing a visual representation of our flow. The streamline is a function of x and y, $\psi(x, y)$. For curves to be tangential to the velocity field the slope of the curve must equal to $\frac{v_{y-direction}}{u_{x-direction}}$: $\frac{dy}{dx} = \frac{v}{u}$. Rearranging this relation, $udy - vdx = 0$. For $\bar{u} = u_i + v_j$ there exists some $\bar{u} = \nabla\psi$ that satisfies $\nabla \cdot \bar{u} = 0$. Therefore $\nabla\psi$ must be:

$$\nabla\psi = \frac{\partial\psi}{\partial y}, \frac{-\partial\psi}{\partial x} \qquad \text{Cartesian}$$
$$\nabla\psi = \frac{1}{r}\frac{\partial\psi}{\partial\theta}, \frac{-\partial\psi}{\partial r} \qquad \text{Polar}$$

These unique values for our velocity field where $u = \frac{\partial\psi}{\partial y}$ and $v = \frac{-\partial\psi}{\partial x}$ allows ψ to satisfy $\nabla^2\psi = 0$. The divergence of $\nabla \cdot \bar{u} = \nabla \cdot \nabla\psi = 0$.

$$\nabla \cdot \bar{u} = \frac{\partial}{\partial x}u + \frac{\partial}{\partial y}v \rightarrow \frac{\partial}{\partial x}\frac{\partial\psi}{\partial y} + \frac{\partial}{\partial y}\frac{-\partial\psi}{\partial y}$$
$$\frac{\partial\psi}{\partial x\partial y} - \frac{\partial\psi}{\partial y\partial x} = 0$$

$\nabla^2\psi = 0$ is satisfied if $u = \frac{\partial\psi}{\partial y}$ and $v = \frac{-\partial\psi}{\partial x}$ therefore $udy - vdx = 0 \rightarrow \frac{\partial\psi}{\partial y}dy - \frac{-\partial\psi}{\partial x}dx = 0$ so $d\psi = 0$. Rewritten $d\psi = udy - vdx$. Integrating $d\psi$ provides our stream function:

$$\int d\psi = \int udy - vdx$$

$$\psi(x, y) = C + \int udy - vdx$$

The following can be written as a complex function:

$$z = x + iy$$
$$\Phi(z) = \varphi(x, y) + i\psi(x, y)$$

ψ is the streamline function and φ is the potential, the gradient streamline if also a solution to the velocity field:

Uniform Potential Flow

Lets analyze a uniform a steady uniform flow in the x-direction, we know:

$$\bar{u} = \frac{\partial u}{\partial x}, \frac{\partial v}{\partial y}$$
$$u = \frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}$$
$$v = \frac{\partial v}{\partial y} = \frac{1}{r}\frac{\partial v}{\partial\theta} = 0$$

If we plug in our potential we find the following:

$$\nabla\varphi = \frac{\partial\varphi}{\partial x}, \frac{\partial\varphi}{\partial y} = \frac{\partial\varphi}{\partial r}, \frac{1}{r}\frac{\partial\varphi}{\partial\theta}$$
$$\frac{\partial\varphi}{\partial y} = 0$$

Since we know $\frac{\partial\varphi}{\partial y} = 0$ we can integrate $\frac{\partial\varphi}{\partial x}$ to find our φ:

$$\varphi = \int\frac{\partial\varphi}{\partial x} = U_x + c \rightarrow \int\frac{\partial\varphi}{\partial x} = U_x$$

$$\varphi = \int \frac{\partial\varphi}{\partial x} = Urcos(\theta) + c \rightarrow \int \frac{\partial\varphi}{\partial x} = Urcos(\theta)$$

Plugging in our streamline function:

$$\nabla\psi = \frac{\partial\psi}{\partial y}, \frac{-\partial\psi}{\partial x} = \frac{1}{r}\frac{\partial\psi}{\partial\theta}, \frac{-\partial\psi}{\partial r}$$
$$\frac{-\partial\psi}{\partial x} = 0$$

Since we know $\frac{-\partial\psi}{dx} = 0$ we can integrate $\frac{\partial\psi}{\partial y}$ to find our ψ:

$$\psi = \int \frac{\partial\psi}{\partial y} = U_y + c \rightarrow \int \frac{\partial\psi}{\partial y} = U_y$$

$$\psi = \int \frac{\partial\psi}{\partial y} = Ursin(\theta) + c \rightarrow \int \frac{\partial\psi}{\partial y} = Ursin(\theta)$$

Sources and Sinks

We first analyze our equations for source/sink streamlines and potentials in polar coordinates:

$$\varphi = \frac{\Lambda}{2\pi} ln(r)$$
$$\varphi = \frac{\Lambda}{2\pi} ln\sqrt{x^2 + y^2}$$
$$\psi = \frac{\Lambda}{2\pi}\theta$$
$$\psi = \frac{\Lambda}{2\pi} tan^{-1}\frac{y}{x}$$

When our Λ is positive we have a source and when Λ is negative it is a sink:

$\Lambda > 0$	Source
$\Lambda < 0$	Sink

Since our potential flows are linear equation the sums of linear equations are solutions:

$$\varphi_3 = \varphi_1 + \varphi_2$$
$$\psi_3 = \psi_1 + \psi_2$$

We can take our solution for our uniform flow and our source, the sum provides a solution for the Rankine Half Body:

$$\psi_3 = \psi_{freestream} + \psi_{source/sink}$$
$$\psi = Ursin(\theta) + \frac{\Lambda}{2\pi}\theta$$

The velocity field from the new stream function come out to be:

$$u = \frac{1}{r}\frac{\partial\psi}{d\theta} = Urcos(\theta) + \frac{\Lambda}{2\pi}$$
$$v = \frac{-\partial\psi}{\partial r} = - Usin(\theta)$$

Our function is two flows interacting, a uniform flow and a source. The source is located at some point α where fluid flows out from. At some point B the uniform and source meet and result in a stagnation point:

$$Urcos(\theta) + \frac{\Lambda}{2\pi} = 0$$
$$- Usin(\theta) = 0$$

$$B(r, \theta) = (\frac{\Lambda}{2\pi U}, \pi)$$

Entering our values for r and θ into our streamline function we get the streamline for our stagnation point:

$$\psi_B = U\frac{\Lambda}{2\pi U} sin(\pi) + \frac{\Lambda}{2\pi}\pi$$
$$\psi_B = \frac{\Lambda}{2}$$

Any streamline outside of ψ_B is considered our flow ψ_F, our ψ_B acts somewhat like the surface of an object.

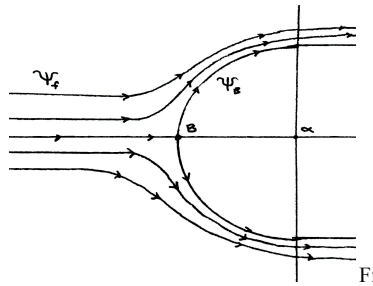

Figure 1

The Rankine Half Body is an infinite object which is unreasonable to assume, we can consider adding a sink of an equal strength to create a finite shape, we know our potential flows are linear allowing us to an infinite sum of flows to create more complex solutions:

$$\psi_4 = \psi_1 + \psi_2 + \psi_3$$
$$\psi_{complex} = \psi_{freestream} + \psi_{source} - \Psi_{sink}$$

We know that Ψ_{sink} will be negative because $\Lambda < 0$ for a sink, our solution becomes:

$$\psi = Ursin\theta + \frac{\Lambda}{2\pi}\theta_1 - \frac{\Lambda}{2\pi}\theta_2$$
$$\psi = Ursin\theta + \frac{\Lambda}{2\pi}(\theta_1 - \theta_2)$$

Our source and sink are set equal (L) distance apart from the origin we use θ_1 and θ_2. We can also rewrite our solution into cartesian coordinates to give us a better understanding:

$$\psi = U_y + \frac{\Lambda}{2\pi}tan^{-1}\frac{y}{x+L} - \frac{\Lambda}{2\pi}tan^{-1}\frac{y}{x-L}$$

Using similar methods to our Rankine Half Body we can found our stagnation points and our vector field for $\bar{u}$ by finding $\nabla\psi$.

Potential Flow Around a Cylinder

To consider the potential flow around a cylinder, what we must do is set the distance between our source and sink as close as possible to 0 but not equal to 0, so $L \to 0$ and our strength of our flow approaches infinity $\Lambda \to \infty$. Let's rewrite our streamline function from our previous chapter:

$$\psi = \frac{\Lambda}{2\pi}\Delta\theta$$
$$\Delta\theta = \frac{Lsin\theta}{r - Lcos\theta}$$
$$\psi = \frac{\Lambda}{2\pi}\frac{Lsin\theta}{r - Lcos\theta}$$

We can say that as $L \to 0$ and $\Lambda \to \infty$ their product will be some constant $\Lambda L = k$:

$$\psi = \frac{k}{2\pi}\frac{sin\theta}{r - Lcos\theta}$$

Since $Lcos\theta \to 0$ because $L \to 0$ our function simplifies to:

$$\psi = \frac{k}{2\pi}\frac{sin\theta}{r}$$

This streamline function is called our doublet, this is a streamline that occurs when our source and sink are infinitesimally close together, we can find the sum of our uniform flow and our doublet and give us the streamlines around our cylinder:

$$\psi = Ursin(\theta) + \frac{k}{2\pi}\frac{sin\theta}{r}$$

You can restructure our equation into this expanded form:

$$\psi = Ur(1 + \frac{k}{2\pi Ur^2})sin\theta$$

Where $R = \sqrt{\frac{k}{2\pi U}}$ and R is the radius of the circle, the following equation:

$$\psi = Ur(1 + \frac{R^2}{r^2})sin\theta$$

Our velocity vectors would be found taking the gradient of our streamline function:

$$u = \frac{1}{r}\frac{\partial\psi}{d\theta} = U(1 - \frac{R^2}{r^2})cos\theta$$

$$v = \frac{-\partial\psi}{\partial r} =- U(1 + \frac{R^2}{r^2})sin\theta$$

Using our velocities we can find our stagnation points:

$$U(1 - \frac{R^2}{r^2})cos\theta = 0$$
$$- U(1 + \frac{R^2}{r^2})sin\theta = 0$$

The stagnation points calculated become:

$$(r, \theta) = (R, 0), (R, \pi)$$
$$\psi = UR(1 + \frac{R^2}{R^2})sin(0, \pi)$$
$$\psi = 0$$

The streamlines (shown on the top half) and pressure field (shown on the bottom half) estimated are the following:

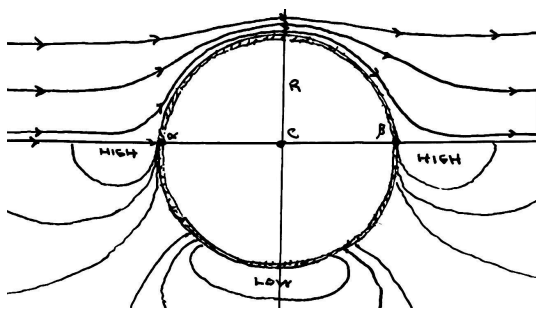

Figure 2

Flow Around a Rotating Cylinder

Considering our potential flow around a cylinder we notice that our pressure gradient is symmetric around our flow, we can conclude there would be no net lift due to this. We need to consider an additional flow Γ or the circulation:

$$\Gamma = \oint_C \bar{u} \cdot ds = \int\int_S (\nabla \times \bar{u}) \cdot ds$$

Our vector field ū for a vortex is the following:

$$\bar{u} = u_r r + v_\theta \theta$$
$$u = 0$$
$$v = \frac{\Gamma}{2\pi r}$$

To find our stream function ψ by integrating our v_θ:

$$v_\theta = \frac{-\partial\psi}{\partial r}$$
$$- \int \frac{\partial\psi}{\partial r} =- \int \frac{\Gamma}{2\pi r}$$
$$\psi = \frac{-\Gamma}{2\pi} ln(r)$$

The curl of our vector field $\nabla \times \bar{u}$ solved for a vortex is the following:

$$\bar{u} = \omega \times r$$
$$\nabla \times \bar{u} = \nabla \times (\omega \times r)$$
$$\nabla \times (\omega \times r) = (\nabla \cdot r)\omega + (r \cdot \nabla)\omega - (\omega \cdot \nabla)r$$
$$\nabla \times \bar{u} = 2\omega$$

Pugging in the curl of our velocity and integrate around our surface which is a cylinder:

$$\Gamma = \iint_s (2\omega) \cdot ds \rightarrow 2\omega \iint_s ds$$

$$\iint_s ds = \pi R^2$$
$$\Gamma = 2\pi R^2 \omega$$

For potential flows we use are are scaled to a factor of 4 the resulting value for Γ:

$$\Gamma = 4\pi R^2 \omega$$

Our streamline function for a rotating cylinder will be the sum of our potential flow around a cylinder and our circulation:

$$\psi_{rotating} = \psi_{cylinder} - \psi_{circulation}$$

Our solution:

$$\psi = Ur(1 + \frac{R^2}{r^2})sin\theta - \frac{\Gamma}{2\pi} ln(\frac{r}{R})$$

We know our value for Γ, where $\Gamma = 4\pi R^2 \omega$:

$$\psi = Ur(1 + \frac{R^2}{r^2})sin\theta - \frac{4\pi R^2 \omega}{2\pi} ln(\frac{r}{R})$$
$$\psi = Ur(1 + \frac{R^2}{r^2})sin\theta - 2\pi R^2 \omega ln(\frac{r}{R})$$

Boundary Layer Theory

$$\nabla \cdot \bar{u} = 0$$
$$\rho[\frac{\partial \bar{u}}{\partial t} + (\bar{u} \cdot \nabla)\bar{u}] = -\nabla P + \mu \nabla^2 \bar{u} + \rho g$$

Imagine a steady, uniform flow moving at some constant rate U in the x-direction where $\bar{u}(x, y, z) = ui + 0j + 0k$. Consider a uniform stream function $\psi = Uy$ where $\nabla\psi = \bar{u} = Ui$. At any point (x, y, z) the flow will move in the i-direction by some velocity U.

Suppose the uniform fluid flow meets a flat plate which is infinitesimally thin. This flat plate extends along the x and z axes at all points where $x \geq 0$, $y = 0$, and z is any value. As the fluid flows over the surface of the flat plate the velocity of the flow directly at the surface is 0, this is called a no slip condition. Thus at $\bar{u}(0, 0, z)$ the fluid and flat plate meet, at $\bar{u}(x \geq 0, 0, z) = 0$ the no slip condition is in effect, and initially when the two meet at $\bar{u}(0, 0, z)$, $\bar{u}(0, y > 0, z) = U$. There is no pressure gradient and it remains constant throughout the system, $\nabla P = 0$. If we do not consider viscous forces this becomes a simple potential flow with boundary conditions where flow at the plate is 0. Taking viscosity into account we intuitively expect the momentum of the fluid to diffuse increasingly into the +y-direction from $y = 0$ the further we go in the +x-direction.

Since flow gets slower as $y \rightarrow 0$ and diffusion along the y-axis gets greater as $x \rightarrow \infty$ we can say there exists some point between $\bar{u} < U$ and $\bar{u} = U$ which we can express as the thickness of the boundary layer δ, any point above δ, $\bar{u} = U$. We expect δ to grow further the flow moves in the x-direction, $\delta(x)$. δ's growth is slowed down and balanced by viscous forces and inertial forces, thus the assumption is $\delta(x)$ will likely grow rapidly initially then the rate decreases as $x \rightarrow \infty$, resembling a square root function. Logically if the free streams strength increases we expect a thinner boundary layer, simply the free streams force is better in resisting the viscous flow at the wall. Accordingly $\delta(x)$ must be scaled by factors of ν and U appropriate to how we theorize they affect the flow:

$$\delta(x) \sim \sqrt{\frac{\nu x}{U}}$$

To increase simplicity imagine we analyze some strip of flow with some infinitesimally small width dx and a length from $y = 0$ to $y \rightarrow \infty$, each point y holds some value $\bar{u}$, where $\bar{u}$ is the velocity. If we assume the same conditions at $t = 0$, $\bar{u}(0, 0) = 0$, and $\bar{u}(0, y > 0) = U$. We expect as time progresses forward, momentum diffuses in the y-direction. This is similar to the One Dimensional Heat Equation where we expect heat to

diffuse over a period of time. For this simplification we neglect the convective, pressure, and gravitational terms and reduce the Navier-Stokes equation to a simplified version modeling something similar to the Heat Equation which we will call the Diffusion Equation:

$$\rho[\frac{\partial u}{\partial t} + (u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z})] = \frac{-\partial P}{\partial x} + \mu(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}) + \rho g x \rightarrow \rho\frac{\partial \bar{u}}{\partial t} = \mu(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}) \rightarrow \rho\frac{\partial \bar{u}}{\partial t} = \mu\frac{\partial^2 u}{\partial y^2} \rightarrow \frac{\partial \bar{u}}{\partial t} = \nu\frac{\partial^2 u}{\partial y^2}$$

Accordingly these specifications are the boundary conditions: At $y = 0$ $u(0,t) = 0$, At $y \rightarrow \infty$ $u(y,t) = U$, At $t = 0$ when $y > 0$, $u(y,0) = U$ and $u(0,0) = 0$. An argument can be made that we are not solely watching energy diffuse in a rod of some fluid, rather as time advances forward we analyze a different spatial point, almost as if tracking a row of particles through a flow. The relationship between time and space can be given by the positional derivative at the freestream, $x'(t) = U$. Altering the equation to $dx = Udt$ and integrating, the solution is $x = Ut$.

Introducing a similarity variable $\eta = y/t^{\beta}$, which is a dimensionless variable which introduces a similarity between ν, y and t, thus $u(y,t) = g(\eta)$. Based on the solution of the one dimensional Heat Equation we know $\beta = 1/2$ and ν must be included. The argument between the relation of space and time must relate to the similarity variable, ie. $u(x,y)$. Meaning $\eta \neq f(y,t)$ but $\eta = f(x,y)$. Since $t = x/U$, the similarity variable transforms:

$$\eta = y\sqrt{\frac{U}{\nu x}}$$

or

$$\eta = \frac{y}{\delta(x)}$$

Transforming variables from t to x there must be a change to the Diffusion Equation where $\partial t \rightarrow \partial x/u$. If there are no fluctuations of velocity at a certain point in space at any moment in time and the Diffusion Equation was studying the velocity of a particle through a flow, we can take the material derivative which is identical to the particle's acceleration:

$$\frac{D\bar{u}}{Dt} = (\bar{u} \cdot \nabla)\bar{u} + \frac{\partial \bar{u}}{\partial t}$$

Since we assume a steady state throughout the volume, the fluid moving through has no change in velocity at any specific point over time. This and our conditions simplify the material derivative and allows us to transform the Diffusion Equation:

$$\frac{\partial \bar{u}}{\partial t} = (\bar{u} \cdot \nabla)\bar{u} \rightarrow \frac{\partial \bar{u}}{\partial t} = \nu\frac{\partial^2 u}{\partial y^2} \rightarrow (\bar{u} \cdot \nabla)\bar{u} = \nu\frac{\partial^2 u}{\partial y^2} \rightarrow u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = \nu\frac{\partial^2 u}{\partial y^2}$$

A solution to this equation must satisfy $\nabla \cdot \bar{u} = 0$. Since the flow is laminar and incompressible a viable solution for mass continuity is the stream function, $\nabla\psi = \bar{u}$. The stream function allows us to find the functions for u and v while complying with conservation laws. We introduce our stream function to the equation above modifying it:

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = \nu\frac{\partial^2 u}{\partial y^2} \rightarrow \frac{\partial \psi}{\partial y}\frac{\partial^2 \psi}{\partial x \partial y} - \frac{\partial \psi}{\partial x}\frac{\partial^2 \psi}{\partial y^2} = \nu\frac{\partial^3 \psi}{\partial y^3}$$

An expanding idea is developing, the nature of the stream equations satisfies the initial hypothesis of how this flow should move and $\nabla \cdot \bar{u} = 0$. The solution of our stream function must include the variable η and be scaled proportionally. We understand the area where our flow will be non uniform is within the boundary layer, so for all $y \leq \delta$. The variation in flow is predominantly influenced by our y variable therefore a solution should predominantly depend on y. Our stream function must have velocity components of only η and remain dimensionally consistent. We can assume a function that contains a coefficient A, variable x, and exponent n. It must contain the variable x because the boundary layer thickness is dependent on the length the flow has traveled, the velocity profile is dependent on the growing boundary layer. There for x must be scaled accurately by some variables A and n. The supposed function must be have solutions in terms of purely η, consider a proposed function:

$$\psi = A\nu x^n U f(\eta)$$

When scaling its possible to assume that u and v that have dimensions of (x,y) corresponding to a spatial length L, in meters for example, and t corresponding to time T, in seconds for example. There for the dimensions for u and v is L/T or meters per second. Variables such as as x, y, ψ, ν , and U all follow similar dimensions of L and T.

$$[x] = L \quad [y] = L \quad [\psi] = \frac{L^2}{T} \quad [\nu] = \frac{L^2}{T} \quad [U] = \frac{L}{T}$$

The stream equation must maintain dimensional consistency with the stream function. The coefficients A and n can be found by checking dimensional consistency:

$$[A][\frac{L^2}{T}][L]^n[\frac{L}{T}] = [\frac{L^2}{T}] \rightarrow [A]\sqrt{[\frac{L^2}{T}][L]^n[\frac{L}{T}]} = [\frac{L^2}{T}]$$

We find that $n = 1$ but for it to be satisfied we must take root of the variables, where the stream function now takes the form, $\psi = [A]\sqrt{\nu x U} f(\eta)$. To find our constant A, we want it to simplify our equation as much as possible. We must begin with finding u and v or $\nabla\psi$:

$$\frac{\partial\psi}{\partial y} = \frac{\partial\psi}{\partial\eta}\frac{\partial\eta}{\partial y} \rightarrow \frac{\partial\psi}{\partial\eta} = [A]\sqrt{\nu x U}f'(\eta)$$
$$\frac{\partial\eta}{\partial y} = \sqrt{\frac{U}{\nu x}}$$
$$\frac{\partial\psi}{\partial y} = [A]\sqrt{\nu x U}f'(\eta)\sqrt{\frac{U}{\nu x}} \rightarrow \frac{\partial\psi}{\partial\eta} = [A]\sqrt{\nu x U}f'(\eta)$$

For the simplest solution, $A = 1$ intuitively makes the most sense. Hence why, $u = Uf'(\eta)$ and $\psi = \sqrt{\nu x U}f(\eta)$. Now solving for our v velocity:

$$\frac{\partial\psi}{\partial x} = \frac{\partial}{\partial x}(\sqrt{\nu x U}f(\eta)) \rightarrow \frac{\partial\psi}{\partial x} = f(\eta)\frac{\partial}{\partial x}(\sqrt{\nu x U}) + (\sqrt{\nu x U})\frac{\partial f(\eta)}{\partial x}$$
$$\frac{\partial}{\partial x}(\sqrt{\nu x U}) = \frac{1}{2}\sqrt{\frac{\nu U}{x}}$$
$$\frac{\partial f(\eta)}{\partial x} = f'(\eta)\frac{\partial\eta}{\partial x} \rightarrow \frac{\partial\eta}{\partial x} = \frac{-1}{2}\eta\frac{1}{x}$$
$$\frac{\partial\psi}{\partial x} = f(\eta)\frac{1}{2}\sqrt{\frac{\nu U}{x}} + (\sqrt{\nu x U})f'(\eta)\frac{-1}{2}\eta\frac{1}{x} \rightarrow \frac{\partial\psi}{\partial x} = \frac{1}{2}\sqrt{\frac{\nu U}{x}}f(\eta) - \frac{1}{2}\sqrt{\frac{\nu U}{x}}\eta f'(\eta)$$
$$v = -\left(\frac{1}{2}\sqrt{\frac{\nu U}{x}}f(\eta) - \frac{1}{2}\sqrt{\frac{\nu U}{x}}\eta f'(\eta)\right) \rightarrow v = \frac{1}{2}\sqrt{\frac{\nu U}{x}}(\eta f'(\eta) - f(\eta))$$

We now have our our u and v velocity vectors in terms of η. We continue by solving for $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$, and $\frac{\partial^2 u}{\partial y^2}$:

$$Uf'(\eta)\frac{\partial}{\partial x}(Uf'(\eta)) + \frac{1}{2}\sqrt{\frac{\nu U}{x}}(\eta f'(\eta) - f(\eta))\frac{\partial}{\partial y}(Uf'(\eta)) = \nu\frac{\partial}{\partial y}\frac{\partial}{\partial y}(Uf'(\eta))$$
$$\frac{\partial u}{\partial x} = Uf''(\eta)\frac{\partial\eta}{\partial x} \rightarrow Uf''(\eta)\frac{-1}{2}\eta\frac{1}{x} \rightarrow \frac{-U\eta}{2x}f''(\eta)$$
$$u\frac{\partial u}{\partial x} = Uf'(\eta)\left(\frac{-U\eta}{2x}f''(\eta)\right)$$
$$\frac{\partial u}{\partial y} = Uf''(\eta)\frac{\partial\eta}{\partial y} \rightarrow Uf''(\eta)\sqrt{\frac{U}{\nu x}}$$
$$v\frac{\partial u}{\partial y} = \frac{1}{2}\sqrt{\frac{\nu U}{x}}(\eta f'(\eta) - f(\eta))Uf''(\eta)\sqrt{\frac{U}{\nu x}} \rightarrow \frac{U^2}{2x}(\eta f'(\eta) - f(\eta))f''(\eta)$$
$$\nu\frac{\partial^2 u}{\partial y^2} = \nu Uf'''(\eta)\frac{U}{\nu x} \rightarrow \frac{U^2}{x}f'''(\eta)$$

The final step in deriving an ODE that we can use methods to solve requires us to plug in our functions of η into our main equation:

$$Uf'(\eta)\left(\frac{-U\eta}{2x}f''(\eta)\right) + \frac{U^2}{2x}(\eta f'(\eta) - f(\eta))f''(\eta) = \frac{U^2}{x}f'''(\eta) \rightarrow f'(\eta)\left(\frac{-U^2}{2x}\eta f''(\eta)\right) + \frac{U^2}{2x}(\eta f'(\eta) - f(\eta))f''(\eta) - \frac{U^2}{x}f'''(\eta) = 0 \rightarrow$$
$$\frac{-1}{2}\eta f'(\eta)f''(\eta) + \frac{1}{2}(\eta f'(\eta) - f(\eta))f''(\eta) - f'''(\eta) = 0 \rightarrow \frac{-1}{2}\eta f'(\eta)f''(\eta) + \frac{1}{2}\eta f'(\eta)f''(\eta) - \frac{1}{2}f'(\eta)f''(\eta) - f'''(\eta) = 0 \rightarrow$$
$$\frac{-1}{2}\eta f'(\eta)f''(\eta) + \frac{1}{2}\eta f'(\eta)f''(\eta) - \frac{1}{2}f'(\eta)f''(\eta) - f'''(\eta) = 0 \rightarrow -\frac{1}{2}f'(\eta)f''(\eta) - f'''(\eta) = 0 \rightarrow \frac{1}{2}f'(\eta)f''(\eta) + f'''(\eta) = 0$$

$$\frac{1}{2}f'f'' + f''' = 0$$